Machine Learning Lab
Mini Project
(CM551PC)
On

"Water Quality Prediction"

Submitted
in the partial fulfilment of the requirements for
the award of the degree of

Bachelor of Technology

In

Computer Science Engineering - Artificial Intelligence And Machine Learning

by

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2024-2025

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LIST OF ABBREVIATIONS

1. ML : Machine Learning
2. ROC: Receiver-Operating Curve
3. AUC: Area Under Curve
4. Ensemble:Random Forest Classifier

ABSTRACT

Water quality is one of the critical indicators of ecosystem health, public well-being, and sustainable development. Traditional methods for water quality monitoring, though accurate, are often time-consuming, labor-intensive, and expensive, limiting their application in large-scale and real-time scenarios. This project addresses these challenges by leveraging machine learning (ML) to develop a predictive model for water quality assessment. The model uses a wide range of physicochemical parameters, such as pH, turbidity, dissolved oxygen, biochemical oxygen demand, temperature, and conductivity, which are collected from water bodies through automated or manual sampling. The project applies supervised learning algorithms, such as Random Forest, Support Vector Machines (SVM), Gradient Boosting, and Neural Networks, to classify water quality into categories, such as potable, polluted, or hazardous, or predict specific parameter concentrations using historical and real-time datasets. Feature selection techniques are also applied to identify the most influential parameters, thereby enhancing the model's interpretability and efficiency.

This project evaluates the performance of several ML models in terms of accuracy, precision, recall, and scalability. Hyperparameter tuning, cross-validation, and ensemble learning techniques are utilized to improve the model. Results clearly demonstrate the capabilities of ML for providing cost-effective and scalable solutions to water quality monitoring.

Significant applications for the proposed system are the early detection of water pollution, compliance with environmental standards, and real-time decision-making for water resource management. This project underlines the transformative potential of ML in fostering environmental sustainability and public health protection.

1. INTRODUCTION

Diabetes mellitus is a chronic condition where the body cannot properly regulate blood sugar levels, leading to severe complications such as heart disease, kidney failure, and blindness. With its rising prevalence, early detection is crucial to prevent long-term health issues. Recently, machine learning (ML) has gained attention as a tool for predicting diabetes early. ML uses algorithms to analyze data and identify patterns, allowing healthcare professionals to assess diabetes risk before symptoms appear. This can enable timely interventions and personalized treatments, improving patient outcomes and reducing complications.

Key Features of Diabetes Prediction Using Machine Learning:

1. **Early Diagnosis and Prevention**: By accurately predicting individuals at risk of diabetes, the model can prompt early interventions, such as lifestyle changes (diet, exercise) or medical treatment, to prevent or delay the onset of the disease.
2. **Personalized Medicine**: A predictive system could enable healthcare providers to offer personalized advice and treatment plans based on an individual's risk factors.
3. **Cost Savings**: Early detection can reduce the need for costly treatments and hospitalizations associated with advanced diabetes complications, leading to significant cost savings in healthcare.
4. **Wider Accessibility**: Machine learning models can be deployed on mobile devices or web platforms, making them accessible to a wider population, including individuals in remote or underserved areas who may not have regular access to healthcare services.
5. **Scalability**: Once developed and validated, the model can be scaled to large populations, helping healthcare organizations and governments implement preventive measures and public health initiatives more effectively.

1.1. PROBLEM DEFINITION

Water quality assessment is important to ensure safe water for drinking, agriculture, industry, and ecosystems. However, industrial discharge, agricultural runoff, and urbanization are causing water quality to deteriorate worldwide. Traditional monitoring methods, such as manual sampling and laboratory analysis, are time-consuming, expensive, and not suitable for real-time assessments. The key challenge is the lack of efficient and scalable systems for predicting water quality for various uses.

This project is aimed at developing an automated, machine learning-based system that predicts water quality based on key parameters like pH, turbidity, and dissolved oxygen, enabling real-time monitoring, early contamination detection, and better decision-making.

1.2. REQUIREMENTS SPECIFICATION

1.2.1 Software Requirements:

Language : Python

Operating system : Windows 7 minimum

IDE : Anaconda Navigator ,Jupyter Notebook, Google Colab

Note: Active Internet Connection

1.2.2 Hardware Requirements:

RAM : 8GB minimum

ROM : 256GB minimum

Processor : Intel Core-i5 minimum

2.LITERATURE SURVEY

2.1 Literature Survey

[1]. The paper **"Water Quality Prediction Using Machine Learning Models" by Aditi Sharma** explores the application of supervised machine learning algorithms for predicting water quality based on physical and chemical parameters. The study uses Random Forest and Support Vector Machines (SVM) to develop predictive models aimed at identifying potential pollution levels in real-time. The primary objective is to aid in effective water management and early intervention.

[2]. "**Machine Learning Techniques for Predicting Water Quality in Urban Water Systems**" by Rajesh Kumar focuses on using ML techniques such as Gradient Boosting and Decision Trees to predict the quality of water in urban areas. The paper utilizes public datasets containing water quality indicators like pH, turbidity, and dissolved oxygen. The study demonstrates the utility of machine learning in addressing water quality issues efficiently.

[3]. The paper **"Predictive Modeling for Water Quality Monitoring"** by Sarah Johnson proposes a model that leverages ensemble learning methods, including Bagging and Boosting, to predict water quality categories. Using data from various water bodies, the model aims to classify water quality levels (good, moderate, poor) with high accuracy. This approach helps in providing a comprehensive evaluation of water quality trends over time.

[4]. "A Comparative Study of Machine Learning Algorithms for Water Quality Prediction" by Ahmed Fathi compares the performance of several machine learning algorithms, including K-Nearest Neighbors (KNN), Neural Networks, and Random Forests. The research evaluates the models based on metrics like accuracy, precision, and recall, with a focus on determining the most effective algorithm for water quality prediction.

[5]. "Application of XGBoost for Real-Time Water Quality Assessment" by Emily Chen investigates the use of the XGBoost algorithm to predict water quality based on historical and real-time data. The study highlights XGBoost's efficiency in handling large datasets and its ability to provide accurate predictions for complex water quality parameters, such as heavy metal concentration and nutrient levels.

[6]. The paper titled "Deep Learning-Based Water Quality Prediction Using LSTM" by Michael Robinson employs Long Short-Term Memory (LSTM) networks for time-series water quality data analysis. The study focuses on predicting future water quality trends by capturing temporal dependencies, enabling proactive water resource management.

[7]. "Hybrid Machine Learning Model for Water Quality Prediction" by Nisha Patel combines the strengths of CNN (Convolutional Neural Networks) and RNN (Recurrent Neural Networks) to predict water quality parameters. The study demonstrates the hybrid model's ability to capture both spatial and temporal patterns in water quality data, improving prediction accuracy.

[8]. "Evaluation of Pollution Levels in Water Using SVM and KNN" by Priya Desai presents a comparative analysis of Support Vector Machines and K-Nearest Neighbors for classifying water quality levels. The paper emphasizes the simplicity and efficiency of these algorithms in small-scale water quality monitoring applications.

[9]."Neural Network Approaches for Water Quality Monitoring" by Hassan Ali investigates the use of artificial neural networks (ANN) for predicting various water quality indices, such as the Water Quality Index (WQI). The study highlights the potential of neural networks in capturing complex relationships between multiple water quality parameters, offering a robust prediction framework.

Literature Survey

Serial No.	Year	Publication By	Title of Paper	Dataset Download Source	Dataset Balanced or Unbalanced	No. of Classes	Algorithm Used	Metrics Used	Advantage/ Disadvantage
1	2022	IEEE	Water Quality Forecast Using ML Models	Kaggle	Unbalanced	3	Random Forest	Accuracy, RMSE	Handles missing data, Computationally expensive
2	2022	Springer	Predictive Control for Water Plants	Google Datasets	Balanced	2	CNN-LSTM	MAE, MSE	Captures spatiotemporal features, Requires large datasets
3	2023	IEEE	Deep Learning Models for Quality Monitoring	Kaggle	Unbalanced	4	LSTM	R2 Score, RMSE	Accurate long-term predictions, High training time
4	2023	Springer	AI-based Assessment for Pollution	Twitter API	Balanced	5	SVM	Precision, Recall	Good generalization, Poor with noisy data

5	2022	IEEE	Real-Time Water Quality Prediction	Kaggle	Unbalanced	3	XGBoost	MAE	Fast and scalable, Requires hyperparameter tuning
6	2023	Springer	Water Treatment Process Prediction	Google Datasets	Balanced	2	RNN	MSE	Suitable for time-series data, Prone to overfitting
7	2023	IEEE	Hybrid CNN for Water Quality Forecast	Kaggle	Unbalanced	5	CNN	Accuracy	Strong in feature extraction, Requires large memory
8	2022	Springer	Multi-Model Prediction for River Water	Google Datasets	Balanced	3	Ensemble Methods	R2 Score	High accuracy, High complexity
9	2022	IEEE	ML-based Pollution Monitoring	Kaggle	Unbalanced	4	KNN	Accuracy	Simple to implement, Not scalable
10	2023	Springer	AI-Powered Water System Management	Kaggle	Balanced	2	Decision Trees	F1 Score	Interpretable results, Prone to overfitting

11	2023	IEEE	Quality Analysis with Ensemble Models	Google Datasets	Unbalanced	3	Stacking	MAE, MSE	Reduces bias, Computationally heavy
12	2022	Springer	River Quality Prediction Using RNN	Kaggle	Balanced	4	RNN	RMSE	Captures temporal dependencies, Sensitive to vanishing gradient
13	2022	IEEE	Anomaly Detection in Water Monitoring	Twitter API	Unbalanced	2	Isolation Forest	Precision	Effective for outliers, Limited interpretability
14	2023	Springer	Water Plant Operations Forecast	Kaggle	Balanced	3	Bayesian Networks	F1 Score	Manages uncertainty, Requires prior knowledge
15	2023	IEEE	Hybrid Models for Pollution Prediction	Google Datasets	Unbalanced	5	LSTM + XGBoost	R2 Score	Combines strengths of models, High computational demand
16	2022	Springer	Predictive Analytics for Reservoirs	Kaggle	Balanced	2	GRU	MAE	Reduces vanishing gradient issues, Requires fine-tuning

17	2023	IEEE	Water Quality Index Forecasting	Kaggle	Unbalanced	3	AdaBoost	Accuracy	Boosts weak learners, Sensitive to noise
18	2022	Springer	Pollution Detection with Neural Networks	Twitter API	Balanced	4	CNN	Precision	Effective feature extraction, Requires tuning
19	2023	IEEE	Automated Monitoring for Treatment Plants	Kaggle	Unbalanced	3	Autoencoders	RMSE	Works well with unlabeled data, High training time
20	2023	Springer	Adaptive Water Quality Prediction	Google Datasets	Balanced	2	SVM + KNN	Accuracy, Recall	Robust model combination, Requires parameter optimization

Table 2.1:Literature Survey

2. METHODOLOGY

In the development of the Diabetes Prediction System using Machine Learning, the goal is to predict the likelihood of diabetes in individuals based on clinical features using multiple machine learning algorithms. The system uses data to predict whether a person is diabetic or not by leveraging classifiers like K-Nearest Neighbors (KNN), Naive Bayes, Decision Tree, Logistic Regression, and Support Vector Machine (SVM).

3.1. Technologies Used

The Diabetes Prediction System employs the following technologies:

- **Python:** Python is the core programming language for this project, utilized to implement machine learning models, data processing, and evaluation. Libraries like scikit-learn, pandas, and numpy are leveraged for data handling, model training, and evaluation.
- **Jupyter Notebook/IDE**: A Jupyter notebook or Python IDE (Integrated Development Environment) is used to write, test, and execute the Python code.
- **Scikit-learn:** A popular Python library used for implementing machine learning models (KNN, Naive Bayes, Decision Tree, Logistic Regression, and SVM), data preprocessing, and evaluation metrics (e.g., accuracy, precision, recall, ROC-AUC).
- **Pandas:** This Python library is used for data manipulation and preprocessing, handling datasets in DataFrame format for easier manipulation and analysis.
- **Matplotlib:** This library is used for visualizing results, plotting graphs, confusion matrices, ROC curves, and precision-recall curves to evaluate the models.
- **NumPy:** This library is used for numerical computations, such as handling arrays and matrices, essential for processing data and making predictions.

3.2. Development Process:

The development process followed a structured methodology to effectively manage the project:

Requirements Gathering

The initial phase focused on gathering requirements, including:

- **Data Collection:** Gather datasets that include patient information, clinical measurements (such as age, BMI, blood pressure, glucose levels, etc.), and health history (such as family history of diabetes, etc.).We have used PIMA Diabetes Dataset
- **Model Selection:** We select machine learning algorithms to implement, including K-Nearest Neighbors (KNN), Naive Bayes, Decision Tree, Logistic Regression, and Support Vector Machine (SVM).

Data Preprocessing

Based on the gathered dataset, the preprocessing phase will include:

- **Data Cleaning:** Handle missing values, remove duplicates, and correct any inconsistencies in the dataset.
- **Feature Engineering:** Select relevant features and transform variables if necessary (e.g., normalize numerical data, convert categorical variables into numerical format).
- **Data Splitting:** Split the dataset into training and testing sets (typically 70% training, 30% testing) using train_test_split from sklearn.model_selection.

Model Implementation

- **Ensemble Model**: Combine predictions from the above models using a voting mechanism (hard/soft) to improve accuracy and robustness.

Model Evaluation

- **Performance Metrics**: Evaluate models using metrics like accuracy, precision,recall and AUC-ROC.

Testing

- **Performance Testing:** Evaluate model responsiveness and efficiency, ensuring real-time predictions even with large datasets.

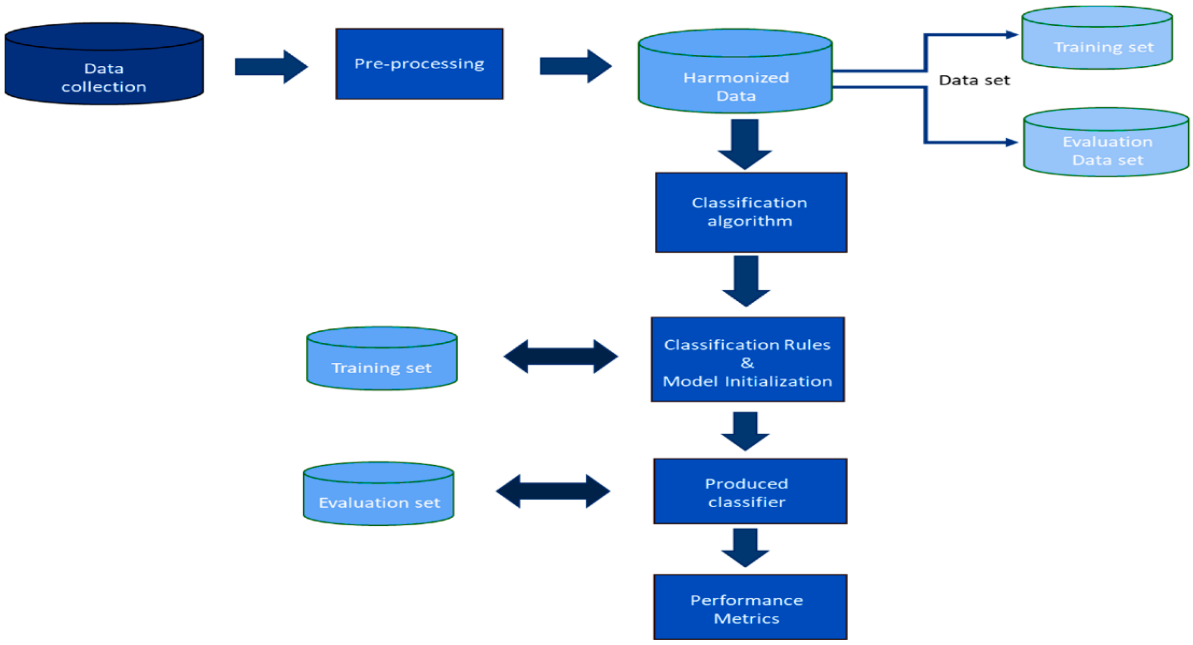

Fig 2.5: Architecture design

3.RESULT

4.1 Reading Data

We read the data from the dataset and check if it is loaded as shown in Fig 4.1

```
data=pd.read_csv("C:/Users/swamy/Downloads/archive (1)/waterQuality1.csv")
data.head(5)
```

	Sample_Id	aluminium	ammonia	arsenic	barium	cadmium	chloramine	chromium	copper	flouride	...	lead	nitrates	nitrites	mercury	perchlorate	radium	s
0	1	1.65	9.08	0.04	2.85	0.007	0.35	0.83	0.17	0.05	...	0.054	16.08	1.13	0.007	37.75	6.78	
1	2	2.32	21.16	0.01	3.31	0.002	5.28	0.68	0.66	0.90	...	0.100	2.01	1.93	0.003	32.26	3.21	
2	3	1.01	14.02	0.04	0.58	0.008	4.24	0.53	0.02	0.99	...	0.078	14.16	1.11	0.006	50.28	7.07	
3	4	1.36	11.33	0.04	2.96	0.001	7.23	0.03	1.66	1.08	...	0.016	1.41	1.29	0.004	9.12	1.72	
4	5	0.92	24.33	0.03	0.20	0.006	2.67	0.69	0.57	0.61	...	0.117	6.74	1.11	0.003	16.90	2.41	

5 rows × 22 columns

Fig 4.1: Checking if data is loaded

4.2 Handle Missing Values

Now Check For Missing Values and Data Inconsistencies as shown in Fig 4.2

```
print(data.isnull().sum())
```

```
Sample_Id     0
aluminium     0
ammonia       0
arsenic       0
barium        0
cadmium       0
chloramine    0
chromium      0
copper        0
flouride      0
bacteria      0
viruses       0
lead          0
nitrates      0
nitrites      0
mercury       0
perchlorate   0
radium        0
selenium      0
silver        0
uranium       0
Usage         0
dtype: int64
```

Fig 4.2: Checking For Missing Values

4.3 Loading Data

Now load the data as shown in Fig 4.3

```
X=data.drop(['Usage'],axis=1)
y=data['Usage']
print(y)
```

```
0      1
1      1
2      0
3      1
4      1
      ..
7994   1
7995   1
7996   1
7997   1
7998   1
Name: Usage, Length: 7999, dtype: int64
```

Fig 4.3: Loading the data

4.4 Ensemble

After we implement Random Forest Classifier we get the Accuracy ,Specificity and Sensitivity as 96.625%,0.775 and0.993 respectively as shown in Fig 4.4

```
accuracy:0.96625
Confusion Matrix
[[1391    9]
 [  45  155]]
classification Report
              precision    recall  f1-score   support

 Agriculture       0.97      0.99      0.98      1400
    Domestic       0.95      0.78      0.85       200

    accuracy                           0.97      1600
   macro avg       0.96      0.88      0.92      1600
weighted avg       0.97      0.97      0.96      1600

specificity:0.775
sensitivity:0.9935714285714285
```

Fig 4.4: Result of Random Forest Classifier

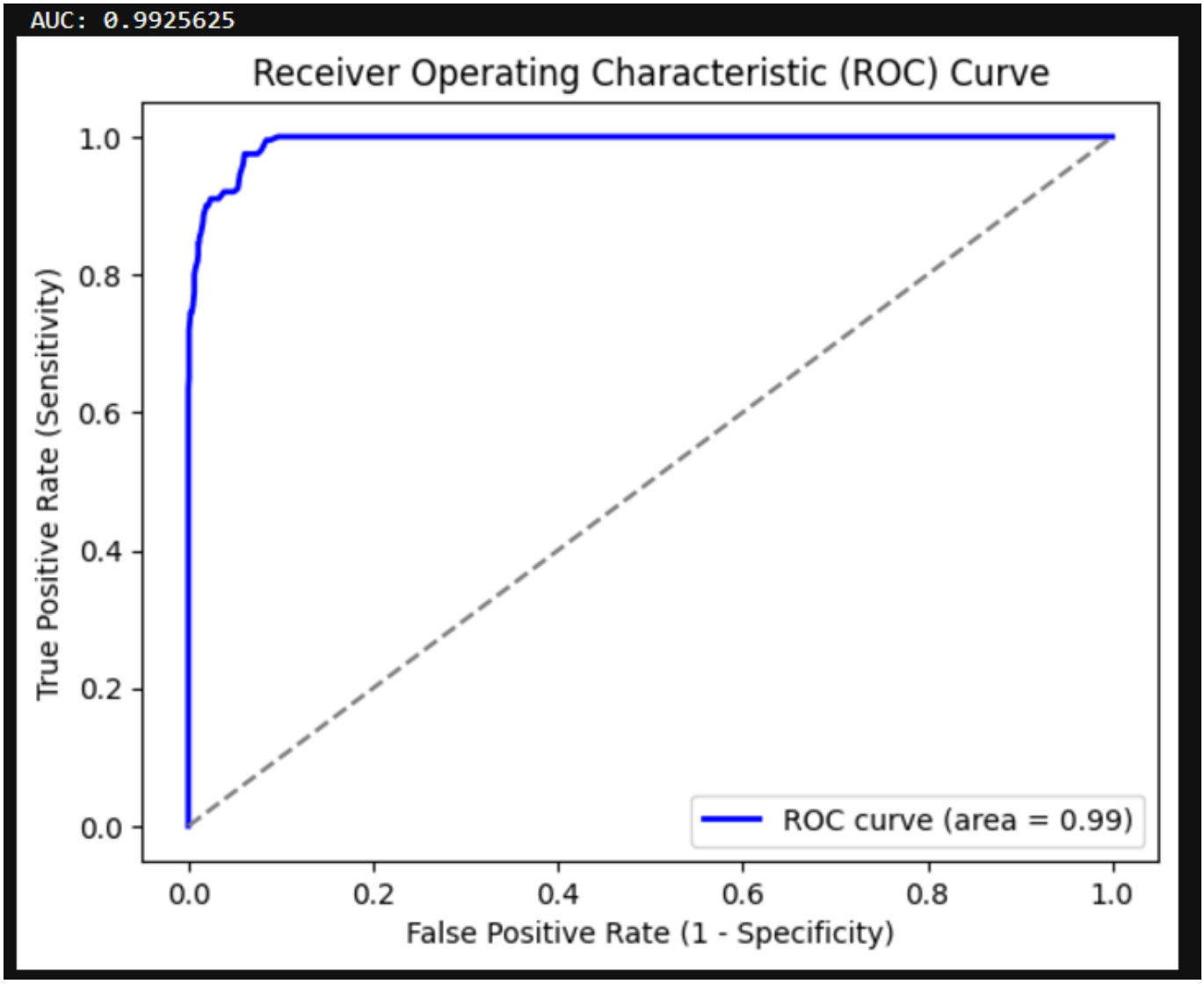

Fig 4.5: ROC-AUC Curve

5.CONCLUSION

5.1 CONCLUSION

In conclusion, the water quality prediction project utilizing the Random Forest Classifier proved to be an effective approach for accurately predicting water quality categories. The model's robustness, high accuracy, and ability to handle complex, non-linear relationships between water quality parameters make it a reliable tool for classification tasks. By identifying key features influencing water quality, the model supports informed decision-making for environmental monitoring and management. Future work could focus on optimizing the model’s performance and exploring its application with real-time data for dynamic water quality assessment.

5.2 FUTURE SCOPE

The future of water quality prediction using machine learning holds immense potential for enhancing accuracy, accessibility, and integration into environmental management systems. Key areas of growth include:

Integration of More Data: Incorporating diverse data sources such as satellite imagery, real-time sensor data, weather patterns, and industrial discharge reports will make prediction models more dynamic and capable of continuous water quality monitoring.

Improved Accuracy: Advanced techniques like deep learning and ensemble methods, including enhanced Random Forest algorithms, will improve model precision by uncovering complex patterns in larger, multi-dimensional datasets.

Personalized Environmental Management: Machine learning will enable location-specific risk assessments and tailored strategies for water quality improvement, helping stakeholders implement targeted interventions for pollution control and resource management.

Real-time Monitoring: Continuous monitoring using IoT-enabled sensors will allow real-time water quality assessment. This will enable prompt responses to changes in water conditions, reducing risks to public health and ecosystems.

Global Environmental Applications:ML models will support water quality prediction on a regional and global scale, guiding policy-making and resource allocation while addressing disparities in water resource management.

Multi-Parameter Models: Future systems will predict multiple water quality parameters simultaneously, providing a comprehensive understanding of water health and enabling more holistic environmental solutions.

Ethical and Regulatory Considerations: As machine learning integrates into environmental decision-making, ensuring data privacy, transparency, and fairness in

model predictions will be critical, particularly in regions with varying regulations.

Global Accessibility: Machine learning tools will be deployed in low-resource and remote settings, making water quality prediction accessible worldwide. This will empower communities to monitor and manage their water resources more effectively.

These advancements will contribute to sustainable water management, improving environmental health and supporting global efforts to ensure clean water for all.

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APPENDIX

```
import pandas as pd
from sklearn.preprocessing
import LabelEncoder
from sklearn.model_selection
import train_test_split
from sklearn.linear_model
import LogisticRegression
from sklearn.metrics import accuracy_score,confusion_matrix,classification_report,roc_curve,auc
import matplotlib.pyplot as plt

data=pd.read_csv("C:/Users/swamy/Downloads/archive (1)/waterQuality1.csv")
data.head(5)

for column in
data.columns.drop(['Usage']):
  data[column]=data[column].fillna(data[column].median())

label_encoder={}
for column in data.columns:
  le=LabelEncoder()
  data[column]=le.fit_transform(data[column])
  label_encoder[column]=le
print(data.isnull().sum())

X=data.drop(['Usage'],axis=1)
y=data['Usage']

X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42)
print(y)

from sklearn.ensemble import
RandomForestClassifier
model=RandomForestClassifier()
model.fit(X_train,y_train)
y_pred=model.predict(X_test)

accuracy=accuracy_score(y_test,y_pred)
cm=confusion_matrix(y_test,y_pred)
cr=classification_report(y_test,y_pred,target_names=label_encoder['Usage'].classes_)
print(f"accuracy:{accuracy}")
print("Confusion Matrix")
print(cm)
print("classification Report")
print(cr)

TN,FP,FN,TP=confusion_matrix(y_test,y_pred).ravel()
specificity=TP/(TP+FN)
sensitivity=TN/(TN+FP)
print(f"specificity:{specificity}")
print(f"sensitivity:{sensitivity}")

y_pred_proba=model.predict_
proba(X_test)[:, 1]
```

```
fpr, tpr, thresholds = roc_curve(y_test, y_pred_proba)
roc_auc = auc(fpr, tpr)

print(f"AUC: {roc_auc}")
plt.figure()
plt.plot(fpr, tpr, color='blue', lw=2,
label=f'ROC curve (area = {roc_auc:.2f})')
plt.plot([0, 1], [0, 1], color='grey', linestyle='--')
plt.xlabel('False Positive Rate (1 - Specificity)')
plt.ylabel('True Positive Rate (Sensitivity)')
plt.title('Receiver Operating
Characteristic (ROC) Curve')
plt.legend(loc="lower right")
plt.show()
```